Lecture 1

Representations of Lie groups and Lie algebras

Objectives. *Part I's central notion — the representation — crosses Lecture 8's bridge. We define representations of matrix Lie groups and of Lie algebras and prove the differentiation theorem that converts the former into the latter, preserving invariant subspaces, intertwiners and unitarity over connected groups; we meet the adjoint representation, under which* $\mathrm{SU}(2)$ *acts on its own algebra by rotations; we determine when algebra representations climb back up — always locally, globally only if their monodromy is trivial; simple connectivity guarantees this, while a nonsimply-connected group requires descent from its universal cover. We prove simple connectivity for* $\mathrm{SU}(2)$ *and exhibit the obstruction for* $\mathrm{U}(1)$*; and we arrive at the physics: Noether's theorem, classical and quantum.*

1.1 Representations, and the circle classified

Part I studied groups through their representations, and the whole doctrine carries over with one new word in the definition.

Definition 1.1 (Representation of a matrix Lie group)**.** A *representation* of a matrix Lie group G on a finite-dimensional complex vector space V is a *continuous* homomorphism $\Pi\colon G \to \mathrm{GL}(V)$. Invariant subspaces, irreducibility, intertwiners $T\Pi(g) = \Pi'(g)T$, and equivalence ($\cong$: an invertible intertwiner) are defined word for word as in Lecture 3.

Definition 1.2 (Representation of a Lie algebra)**.** A *representation* of a real Lie algebra $\mathfrak{g}$ on V is a real-linear map $\pi\colon \mathfrak{g} \to \mathfrak{gl}(V)$, the algebra of all linear operators on V under the commutator, with $\pi([X,Y]) = [\pi(X), \pi(Y)]$. The same vocabulary applies: W is invariant if $\pi(X)W \subseteq W$ for all X; intertwiners obey $T\pi(X) = \pi'(X)T$.

Continuity is the one genuinely new requirement, and Lecture 8 has already shown what it buys: by the one-parameter theorem, continuous homomorphisms are secretly analytic. This lecture exploits that miracle systematically.

Remark 1.3 (Scope at the Lie-group transition)**.** Until Section 1.5.2, every representation in this lecture is continuous and finite-dimensional, and every derived Lie-algebra operator is bounded. Schur's lemma transfers as finite-dimensional linear algebra; Maschke and the finite-group trace/counting formulae transfer only when a compact-group analogue is explicitly proved in Lecture 12. The quantum section changes category to strongly continuous unitary Hilbert-space representations and imports the corresponding unbounded-generator theorem.

Example 1.4 (First examples, in pairs). Let $G \subseteq \mathrm{GL}(n, \mathbb{C})$ be a matrix Lie group with algebra $\mathfrak{g}$. *Trivial:* $\Pi(g) = \mathbb{1}$ on any V; $\pi(X) = 0$. *Defining:* $\Pi(A) = A$ on $\mathbb{C}^n$; $\pi(X) = X$. *Determinant:* $\Pi(A) = \det A$ on $\mathbb{C}$; $\pi(X) = \operatorname{tr} X$ — a representation since $\operatorname{tr}[X, Y] = 0$. In each pair the algebra entry is the derivative of the group entry at the identity; for the determinant this is exactly Lecture 8's identity $\det e^{tX} = e^{t \operatorname{tr} X}$. That the pairing is a theorem, not a coincidence, is Section 1.2's business.

The first classification theorem of Part II concerns the smallest infinite group in the zoo, and its answer is the prototype of every quantum number to come.

Theorem 1.5 (The irreducible representations of U(1)). *Every irreducible representation of* $\mathrm{U}(1)$ *is one-dimensional, and the complete list, up to equivalence, is*

$$\chi_n\big(e^{\mathrm{i}\theta}\big) = e^{\mathrm{i}n\theta}, \qquad n \in \mathbb{Z}. \tag{1.1}$$

Distinct n *give inequivalent representations, and every* χ_n *is unitary.*

Proof. Schur's lemma is a statement of pure linear algebra — an eigenspace of an operator commuting with the representation is invariant — so Lecture 3's proof applies verbatim to any group: in an irreducible complex representation, anything commuting with the whole image is a scalar. Since $\mathrm{U}(1)$ is abelian, each $\Pi(z)$ commutes with the image, hence $\Pi(z) = c(z)\mathbb{1}$; every line in V is then invariant, and irreducibility forces $\dim V = 1$.

A one-dimensional representation is a continuous homomorphism into $\mathrm{GL}(1, \mathbb{C}) = \mathbb{C}^\times$. Composing with $\theta \mapsto e^{\mathrm{i}\theta}$ gives a continuous homomorphism $\mathbb{R} \to \mathbb{C}^\times$, which by Lecture 8's one-parameter theorem is $\theta \mapsto e^{\theta a}$ for a unique $a \in \mathbb{C}$. Descending to the circle is the constraint: $e^{2\pi a} = \Pi(1) = \mathbb{1}$ forces $2\pi a \in 2\pi\mathrm{i}\mathbb{Z}$, i.e. $a = \mathrm{i}n$ with $n \in \mathbb{Z}$. One-dimensional representations are equivalent only if equal, and the functions $e^{\mathrm{i}n\theta}$ are distinct; each has modulus one, hence is unitary. ∎

The integer was not imposed; it was extracted from the requirement that a continuous assignment return to its starting value after one turn of the circle. This is the course's first *quantisation by topology*, and the mechanism — a closed loop in the group forcing integrality on its representations — runs deeper than it yet appears: Section 1.4 will measure the loop itself, and Section 1.5 will recognise (1.1) inside the quantum theory of angular momentum. (The χ_n are also the Fourier modes: that Fourier analysis *is* the representation theory of $\mathrm{U}(1)$ is a slogan Lecture 12 makes precise.)

1.2 Differentiating a representation

Lecture 8 built the bridge between a group and its algebra; we now send representations across it, downhill first. The following theorem is the workhorse of Part II: every later lecture leans on it, usually without comment.

Theorem 1.6 (Differentiation of representations). *Let* Π *be a representation of a matrix Lie group* G *on* V*, and* $\mathfrak{g} = \mathrm{Lie}(G)$.

(i) *For every* $X \in \mathfrak{g}$ *there is a unique* $\pi(X) \in \mathfrak{gl}(V)$ *with*

$$\Pi\big(e^{tX}\big) = e^{t\pi(X)} \qquad \textit{for all } t \in \mathbb{R}, \tag{1.2}$$

namely $\pi(X) = \frac{\mathrm{d}}{\mathrm{d}t}\,\Pi(e^{tX})\big|_{t=0}$.

(ii) *π is a representation of $\mathfrak{g}$, the unique linear map with $\Pi(e^X) = e^{\pi(X)}$; moreover $\pi(gXg^{-1}) = \Pi(g)\,\pi(X)\,\Pi(g)^{-1}$ for all $g \in G$.*

If, in addition, G is connected:

(iii) *a subspace $W \subseteq V$ is invariant under all $\Pi(g)$ if and only if under all $\pi(X)$; hence Π is irreducible iff π is;*

(iv) *T intertwines Π, Π' iff T intertwines π, π'; hence $\Pi \cong \Pi'$ iff $\pi \cong \pi'$;*

(v) *Π is unitary for an inner product on V iff every $\pi(X)$ is anti-Hermitian for it.*

Proof. (i) $t \mapsto \Pi(e^{tX})$ is a continuous homomorphism $\mathbb{R} \to \mathrm{GL}(V)$; Lecture 8's one-parameter theorem provides a unique generator.

(ii) Homogeneity $\pi(sX) = s\pi(X)$ is the uniqueness in (i). For additivity, push the Lie product formula through the continuous homomorphism Π:

$$\Pi\big(e^{t(X+Y)}\big) = \lim_{m\to\infty}\Big(\Pi\big(e^{tX/m}\big)\Pi\big(e^{tY/m}\big)\Big)^m = \lim_{m\to\infty}\Big(e^{t\pi(X)/m}\,e^{t\pi(Y)/m}\Big)^m = e^{t(\pi(X)+\pi(Y))},$$

and apply uniqueness. Equivariance gives

$$\begin{aligned}\Pi(e^{t\,gXg^{-1}}) &= \Pi(g)\,e^{t\pi(X)}\,\Pi(g)^{-1}\\ &= \exp\big(t\,\Pi(g)\pi(X)\Pi(g)^{-1}\big),\end{aligned}$$

and uniqueness applies again. Now put $g = e^{sY}$ and differentiate at $s = 0$: the curve $s \mapsto e^{sY}Xe^{-sY}$ has velocity $[Y, X]$ and lies in the closed subspace $\mathfrak{g}$, on which the linear map π is continuous, so the left side differentiates to $\pi([Y, X])$; the right side, by the product rule, to $[\pi(Y), \pi(X)]$. So π preserves brackets. Uniqueness as stated: a linear $\tilde{\pi}$ with $\Pi(e^X) = e^{\tilde{\pi}(X)}$, applied to tX, satisfies (1.2), so $\tilde{\pi} = \pi$.

(iii) If $\Pi(e^{tX})w \in W$ for all t, then $\pi(X)w = \lim_{t\to 0}\big(\Pi(e^{tX})w - w\big)/t$ lies in the closed subspace W. Conversely, if $\pi(X)W \subseteq W$ then every partial sum of $e^{t\pi(X)}w$ lies in W, hence the limit does, so $\Pi(e^X)$ preserves W; and a connected G consists of finite products of exponentials (Lecture 8), so every $\Pi(g)$ preserves W. The irreducibility claim follows.

(iv) and (v) are the same two-step pattern. Differentiate $T\,\Pi(e^{tX}) = \Pi'(e^{tX})\,T$ at $t = 0$ to get $T\pi(X) = \pi'(X)T$; conversely multiply the exponential series by T termwise and use connectedness. For (v), differentiating $\big\langle \Pi(e^{tX})v, \Pi(e^{tX})w\big\rangle = \langle v, w\rangle$ at $t = 0$ gives $\pi(X)^\dagger = -\pi(X)$; conversely anti-Hermitian generators exponentiate to unitaries (Lecture 8's $\mathfrak{u}(n)$ computation, transplanted), and products of unitaries are unitary. ■

Three remarks. Everything above holds verbatim for representations on *real* vector spaces; we will need that exactly once, in the next section. Second, mere continuity of Π has again upgraded itself to analyticity — Lecture 8's miracle, propagating. Third, connectedness is not decoration: on $\mathrm{O}(n)$ the determinant representation and the trivial representation both differentiate to $\pi = 0$ (antisymmetric matrices are traceless), yet they are not equivalent. The derivative sees only the identity component, exactly as Lecture 8 warned.

Downhill traffic is now unrestricted. The uphill direction is Section 1.4; first, every group ships with one representation built in.

1.3 The adjoint representation

A matrix Lie group acts on its own Lie algebra: by Lecture 8's closure theorem, $gXg^{-1} \in \mathfrak{g}$ whenever $g \in G$ and $X \in \mathfrak{g}$.

Definition 1.7 (Adjoint representation). For $g \in G$ define $\mathrm{Ad}_g : \mathfrak{g} \to \mathfrak{g}$, $\mathrm{Ad}_g X = gXg^{-1}$. Each Ad_g is linear and invertible (inverse $\mathrm{Ad}_{g^{-1}}$), $\mathrm{Ad}_{gh} = \mathrm{Ad}_g \mathrm{Ad}_h$, and the entries of Ad_g are polynomial in those of g and g^{-1}, hence continuous in g: the *adjoint representation* $\mathrm{Ad}: G \to \mathrm{GL}(\mathfrak{g})$, a representation of G on the real vector space $\mathfrak{g}$.

Proposition 1.8 (The derivative of Ad is ad). *The derivative of* Ad *is* $\mathrm{ad}_X = [X, \cdot\,]$*; explicitly* $\mathrm{Ad}(e^X) = e^{\mathrm{ad}_X}$*, and* $\mathrm{ad}: \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ *is a representation of the algebra on itself. Written out, the statement that* ad *preserves brackets,* $\mathrm{ad}_{[X,Y]} = [\mathrm{ad}_X, \mathrm{ad}_Y]$*, is the Jacobi identity.*

Proof. The curve $t \mapsto \mathrm{Ad}(e^{tX})Y = e^{tX}Ye^{-tX}$ has velocity $[X,Y]$ at $t = 0$ (Lecture 8's computation), so Theorem 1.6 applies with $\pi = \mathrm{ad}$ and yields both claims. For the last: $\mathrm{ad}_{[X,Y]} Z = [[X,Y],Z]$, while $[\mathrm{ad}_X, \mathrm{ad}_Y]Z = [X,[Y,Z]] - [Y,[X,Z]]$; their equality rearranges to $[[X,Y],Z] + [[Y,Z],X] + [[Z,X],Y] = 0$. The Jacobi identity is not an axiom chosen for elegance; it is the statement that every Lie algebra represents itself. ■

Example 1.9 (The adjoint representation of SU(2): rotations). Take $G = \mathrm{SU}(2)$ with Lecture 8's basis $X_a = -\frac{\mathrm{i}}{2}\sigma_a$ of $\mathfrak{su}(2)$, $[X_j, X_k] = \epsilon_{jkl}X_l$. Equip $\mathfrak{su}(2)$ with the real inner product $\langle A, B\rangle = -2\,\mathrm{tr}(AB)$, which makes $\{X_a\}$ orthonormal: $-2\,\mathrm{tr}(X_jX_k) = \frac{1}{2}\,\mathrm{tr}(\sigma_j\sigma_k) = \delta_{jk}$. Cyclicity of the trace gives $\langle \mathrm{Ad}_U A, \mathrm{Ad}_U B\rangle = \langle A, B\rangle$, so in this basis every Ad_U is an orthogonal 3×3 matrix; its determinant is continuous in U, equals 1 at $U = \mathbb{1}$, and SU(2) is connected (it is the sphere S^3), so

$$\mathrm{Ad}: \mathrm{SU}(2) \longrightarrow \mathrm{SO}(3).$$

Which rotations? In the basis (X_1, X_2, X_3) the operator ad_{X_3} sends $X_1 \mapsto X_2$, $X_2 \mapsto -X_1$, $X_3 \mapsto 0$: its matrix is L_3, Lecture 8's generator of rotations about the third axis — and likewise $\mathrm{ad}_{X_a} = L_a$ for each a. *The isomorphism* $X_a \mapsto L_a$ *of Lecture 8 is the adjoint representation in disguise.* Consequently

$$\mathrm{Ad}\big(e^{-\mathrm{i}\frac{\theta}{2}\sigma_3}\big) = e^{\theta\,\mathrm{ad}_{X_3}} = e^{\theta L_3} = R_z(\theta), \tag{1.3}$$

and the same for any axis. The halved angle of Lecture 8 now has a job: as θ runs from 0 to 2π the rotation returns to $\mathbb{1}$, but the SU(2) element arrives at $e^{-\mathrm{i}\pi\sigma_3} = -\mathbb{1}$. So $\mathrm{Ad}(-\mathbb{1}) = \mathrm{Ad}(\mathbb{1})$: the map is two-to-one along this circle, and $\pm\mathbb{1}$ lies in its kernel. That the kernel is *exactly* $\{\pm\mathbb{1}\}$, that Ad is onto SO(3), and what this means for spin, is Lecture 11's story; the present lecture supplies its tools.

1.4 Climbing back: loops, simple connectivity, and lifting

Descending from Π to π loses nothing over a connected group; the natural hope is that ascending loses nothing either. Locally the hope is justified: near the identity, $\Pi(e^X) = e^{\pi(X)}$ *defines* a candidate Π from π. The question is global — whether the local definitions, carried across the whole group, agree with one another — and the simplest infinite group in the zoo already answers it in the negative.

Example 1.10 (A failed ascent). The algebra $\mathfrak{u}(1) = \mathrm{i}\mathbb{R}$ is one-dimensional and abelian, so for every $\lambda \in \mathbb{R}$ the prescription

$$\pi_\lambda(\mathrm{i}\theta) = \mathrm{i}\lambda\theta$$

is a representation of $\mathfrak{u}(1)$ on $\mathbb{C}$. If π_λ came from a representation Π of U(1), the bridge (1.2) would force $\Pi(e^{\mathrm{i}\theta}) = e^{\mathrm{i}\lambda\theta}$ — and the left-hand side returns to its starting value when θ advances by 2π, so the right-hand side must: $e^{2\pi\mathrm{i}\lambda} = 1$, i.e. $\lambda \in \mathbb{Z}$. For integer λ the formula is Theorem 1.5's χ_λ, and all is well; for any other λ there is no representation of

$\mathrm{U}(1)$ whose derivative is π_λ. The failure is instructive twice over. First, it is not local: $e^{\mathrm{i}\lambda\theta}$ is a flawless homomorphism on any small arc, and for $\lambda = \frac{1}{2}$ the attempt to continue it once around the circle returns not to 1 but to -1 — off by a sign that no local repair can remove. (Half-integers, thus expelled, will be readmitted with honours in Lectures 10 and 11.) Second, the failure is the circle's fault, not the formula's: the same π_λ integrates perfectly to a representation $t \mapsto e^{\mathrm{i}\lambda t}$ of the group $(\mathbb{R}, +)$, which traverses the same one-dimensional algebra but contains no loop. The obstruction to climbing lives in the closed loops of the group.

The diagnosis needs a vocabulary, and metric spaces provide all of it.

Definition 1.11 (Loops, homotopy, simple connectivity)**.** Let M be a metric space and $m_0 \in M$. A *loop* at m_0 is a continuous map $\gamma\colon [0,1] \to M$ with $\gamma(0) = \gamma(1) = m_0$. Two loops γ_0, γ_1 at m_0 are *homotopic* if there is a continuous $H\colon [0,1] \times [0,1] \to M$ with $H(t,0) = \gamma_0(t)$, $H(t,1) = \gamma_1(t)$, and $H(0,u) = H(1,u) = m_0$ for all u — a continuous interpolation through loops, basepoint pinned. Running one homotopy for $u \in [0, \frac{1}{2}]$ and a second for $u \in [\frac{1}{2}, 1]$ shows homotopy to be an equivalence relation. A loop is *contractible* if it is homotopic to the constant loop at m_0, and M is *simply connected* if it is path-connected and every loop, at every basepoint, is contractible.

S^2: every loop contracts U(1): this one cannot

Figure 1.1: Two topologies for loops. On the sphere every loop shrinks through intermediate loops (dashed) to its basepoint; the same holds on S^3, though that picture resists printing. On the circle a loop that winds twice — drawn slightly off the circle so that both passes are visible — can never contract: its winding number is pinned.

Two computations now carry the whole section: the sphere S^3 is simply connected, the circle is not. Lecture 8 identified $\mathrm{SU}(2)$ with the unit sphere $S^3 \subset \mathbb{R}^4$, so the first computation is a statement about the group that owns the next two lectures.

Theorem 1.12 (SU(2) is simply connected)**.** $\mathrm{SU}(2) \cong S^3$ *is simply connected. So, by the same proof, is every sphere* S^n *with* $n \geq 2$.

Proof. Let γ be a loop at $m_0 \in S^3$, the unit sphere in $\mathbb{R}^4$ with the Euclidean inner product $\langle \cdot, \cdot \rangle$. The proof straightens γ into a spherical polygon, slides the polygon off some point, and contracts through the resulting puncture.

Step 1: straighten. Being continuous on the compact $[0,1]$, γ is uniformly continuous; choose a partition $0 = t_0 < t_1 < \cdots < t_N = 1$ with $|\gamma(t) - \gamma(t_i)| < \frac{1}{2}$ for all $t \in [t_i, t_{i+1}]$. Let $q\colon [0,1] \to \mathbb{R}^4$ be the piecewise linear path with $q(t_i) = \gamma(t_i)$, interpolating linearly between consecutive sample points. Fix $t \in [t_i, t_{i+1}]$ and $u \in [0,1]$: the point $(1-u)\gamma(t) + u\, q(t)$ is a convex combination of points of the ball of radius $\frac{1}{2}$ about the unit vector $\gamma(t_i)$ — balls are convex, and both $\gamma(t)$ and $q(t)$ lie in this one — so its norm exceeds $\|\gamma(t_i)\| - \frac{1}{2} = \frac{1}{2}$.

Normalising is therefore legal:

$$H(t,u) = \frac{(1-u)\,\gamma(t) + u\,q(t)}{\|(1-u)\,\gamma(t) + u\,q(t)\|}$$

is continuous, equals γ at $u = 0$, fixes the basepoint (at $t \in \{0,1\}$ both paths sit at m_0), and at $u = 1$ delivers the *spherical polygon* $p = q/\|q\|$. Thus γ is homotopic to p.

Step 2: miss a point. On $[t_i, t_{i+1}]$ the path q stays in the subspace $W_i = \operatorname{span}\{\gamma(t_i), \gamma(t_{i+1})\}$, of dimension at most two; hence p maps that interval into $C_i = S^3 \cap W_i$, a great circle or an antipodal pair. Each C_i is closed and has empty interior in S^3: given $x \in C_i$, pick a unit vector v orthogonal to W_i (possible, since $\dim W_i^\perp \geq 2$); then $(x+\delta v)/\sqrt{1+\delta^2}$ lies on S^3, lies outside W_i, and approaches x as $\delta \to 0$, so no neighbourhood of x is contained in C_i. Now cut a ball down finitely many times: take any closed ball $\bar{B}_0 \subset S^3$; since C_0 is closed with empty interior, the interior of $\bar{B}_0$ contains a point off C_0, hence a closed ball $\bar{B}_1 \subset \bar{B}_0$ disjoint from C_0; repeating through $C_1, \dots, C_{N-1}$ produces $\bar{B}_N$ disjoint from every C_i. Any $q^* \in \bar{B}_N$ avoids the image of p.

Step 3: contract through the puncture. Stereographic projection from q^*,

$$\sigma(x) = \frac{x - \langle x, q^*\rangle\, q^*}{1 - \langle x, q^*\rangle}, \qquad \sigma^{-1}(y) = \frac{2y + \left(|y|^2 - 1\right)q^*}{|y|^2 + 1},$$

maps $S^3 \setminus \{q^*\}$ to the hyperplane $(q^*)^\perp \cong \mathbb{R}^3$ and back; both formulas are rational with nonvanishing denominators on their domains, hence continuous, and a direct computation confirms they are mutually inverse. So $S^3 \setminus \{q^*\}$ is homeomorphic to $\mathbb{R}^3$, where straight lines contract everything:

$$K(t,u) = \sigma^{-1}\Big((1-u)\,\sigma\big(p(t)\big) + u\,\sigma(m_0)\Big)$$

is a homotopy, with basepoint fixed, from p to the constant loop at m_0. By transitivity γ is contractible.

For S^n with $n \geq 2$ the argument is verbatim: Step 2 needs only $\dim W_i^\perp \geq 1$, i.e. $n+1 > 2$. And there the induction on hypotheses stops: for $n = 1$ a spherical polygon can cover the whole circle, Step 2 has nothing to miss, and — as the next proposition shows — the conclusion is genuinely false. ∎

Proposition 1.13 (U(1) is not simply connected)**.** *The loop $\gamma_1(t) = e^{2\pi i t}$ is not contractible in* $\mathrm{U}(1)$.

Proof. The instrument is a counter that no homotopy can move. First, *every* loop γ in $\mathrm{U}(1)$ lifts: there is a continuous $\vartheta\colon [0,1] \to \mathbb{R}$ with $\gamma(t) = e^{i\vartheta(t)}$. (Partition $[0,1]$ finely enough that $|\gamma(t) - \gamma(t_i)| < 2$ on each subinterval; on the unit circle the parallelogram law gives $|\gamma - \gamma'|^2 + |\gamma + \gamma'|^2 = 4$, so the ratio $\gamma(t)/\gamma(t_i)$ never equals -1 and carries a continuous principal argument; patch these arguments interval by interval.) Define the *winding number* $w(\gamma) = \big(\vartheta(1) - \vartheta(0)\big)/2\pi$, an integer because $\gamma(1) = \gamma(0)$, and independent of the lift, because two lifts differ by a continuous — hence constant — element of $2\pi\mathbb{Z}$.

Three properties make w rigid. Lifts add under pointwise products, so $w(\gamma\gamma') = w(\gamma) + w(\gamma')$. If $|\gamma(t) - \gamma'(t)| < 2$ for all t and both loops share a basepoint, then $\delta = \gamma'/\gamma$ is a loop at 1 avoiding -1; its lift from 0 can never reach $\pm\pi$ without δ touching -1, so it stays in $(-\pi, \pi)$, forcing $w(\delta) \in (-\frac{1}{2}, \frac{1}{2}) \cap \mathbb{Z} = \{0\}$ and hence $w(\gamma') = w(\gamma)$. Finally, a homotopy H is uniformly continuous on the compact square, so loops $H(\cdot, u)$ and $H(\cdot, u')$ are uniformly within 2 of each other once $|u - u'|$ is small: the function $u \mapsto w\big(H(\cdot, u)\big)$ is locally constant on $[0, 1]$, hence constant.

Now count: γ_1 lifts to $\vartheta(t) = 2\pi t$, so $w(\gamma_1) = 1$, while every constant loop has winding 0. No homotopy connects them. (The patching and the three properties are rehearsed in full on Sheet 9.) ∎

The counter explains Example 1.10 exactly. Under χ_n the fundamental loop γ_1 maps to the loop $e^{2\pi i n t}$, of winding n: the quantum number of Theorem 1.5 literally counts how many times the representation wraps the circle around itself, and a hypothetical $\lambda \notin \mathbb{Z}$ would demand a fractional wrap that continuity forbids.

With the obstruction measured, we can state what happens when it is absent. Two theorems are imported here — stated precisely, used as black boxes, their proofs belonging to a longer course — and both debts carry repayment schedules.

Theorem 1.14 (Baker–Campbell–Hausdorff; imported). *For all X, Y in a sufficiently small ball about 0 in $M_n(\mathbb{C})$,*

$$\log\big(e^X e^Y\big) = X + Y + \tfrac{1}{2}[X, Y] + \tfrac{1}{12}\Big([X, [X, Y]] + [Y, [Y, X]]\Big) + \cdots, \tag{1.4}$$

a convergent series in which every unwritten term is an iterated bracket of X's and Y's with a universal rational coefficient.

Reference for the omitted proof: Hall, Ch. 5.

Lecture 8 promised that the bracket is the germ of the multiplication; this is the precise sense. Near the identity the product of two exponentials is again an exponential, of an element computable from X, Y and brackets alone: *the algebra knows the local group law*. Sheet 9 extracts the displayed terms from the exponential series by hand; Lecture 13 repays a larger instalment by proving the formula *exactly* — the series terminates — for the nilpotent algebra of the Heisenberg group.

Theorem 1.15 (Lifting homomorphisms; imported). *Let G and H be matrix Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, let G be connected and simply connected, and let $\varphi\colon \mathfrak{g} \to \mathfrak{h}$ be a Lie algebra homomorphism. Then there is a unique continuous homomorphism $\Phi\colon G \to H$ with*

$$\Phi\big(e^X\big) = e^{\varphi(X)} \qquad \text{for all } X \in \mathfrak{g}.$$

Reference for the omitted proof: Hall, Ch. 5.

The uniqueness is ours already: two such Φ agree near the identity, hence on the finite products of exponentials that fill a connected group (Lecture 8). It is the existence that costs, and the cost is exactly one homotopy.

Remark 1.16 (Why simple connectivity is the right currency). The construction behind Theorem 1.15 deserves its sketch, because Example 1.10 is precisely this construction failing. To define $\Phi(g)$, join 1 to g by a path and chop the path so finely that each step is e^{X_k} for a small $X_k \in \mathfrak{g}$; declare $\Phi(g)$ to be the ordered product of the $e^{\varphi(X_k)}$. Refining the subdivision does not change the product — that is Baker–Campbell–Hausdorff, applied on both sides of φ, which respects every term of (1.4) because it respects brackets. The product therefore depends only on the homotopy class of the chosen path; when every two paths are homotopic — guaranteed by simple connectivity, since a loop is exactly a pair of paths to compare — the recipe is well defined, and multiplicativity follows by concatenating paths. On $\mathrm{U}(1)$ the recipe, run for π_λ, returns $e^{i\lambda\theta}$ along the short way to $e^{i\theta}$ and $e^{i\lambda(\theta+2\pi)}$ along the way that detours once around: the mismatch $e^{2\pi i\lambda}$ is the *monodromy* of the loop, invisible exactly when $\lambda \in \mathbb{Z}$.

Equivalently, let $q\colon \widetilde{G} \to G$ be the universal covering homomorphism of a connected Lie group. Every finite-dimensional Lie-algebra representation integrates uniquely to a representation $\widetilde{\Pi}$ of the simply connected group $\widetilde{G}$; it descends to a representation of G exactly when $\widetilde{\Pi}(k) = \mathbb{1}$ for every $k \in \ker q$. Indeed this condition makes $\widetilde{\Pi}$ constant on the fibres $\widetilde{g}\ker q$ and hence defines $\Pi(q(\widetilde{g})) = \widetilde{\Pi}(\widetilde{g})$. For $q\colon \mathbb{R} \to \mathrm{U}(1)$, $q(t) = e^{\mathrm{i}t}$, the kernel is $2\pi\mathbb{Z}$, and this criterion is precisely $e^{2\pi i\lambda} = 1$. Thus simple connectivity is a sufficient guarantee for every algebra representation, not a necessary condition for a particular one to integrate.

Corollary 1.17 (The dictionary, perfected)**.** *Let G be connected and simply connected. Then differentiation $\Pi \mapsto \pi$ is a bijection from representations of G on V to representations of $\mathfrak{g}$ on V.*

Proof. Injectivity is connectedness: Π is recovered from π on exponentials by (1.2), and exponentials generate G. For surjectivity, $\mathrm{GL}(V)$ is a matrix Lie group with Lie algebra $\mathfrak{gl}(V)$ (Lecture 8), so Theorem 1.15 applied to $\varphi = \pi$ produces a continuous homomorphism $\Phi\colon G \to \mathrm{GL}(V)$ — a representation — whose derivative is π. ∎

group	simply connected?	which π integrate
$(\mathbb{R}, +)$	yes: it is convex	all
$\mathrm{U}(1)$	no: winding	those with $e^{2\pi\,\pi(\mathrm{i})} = \mathbb{1}$
$\mathrm{SU}(2)$	yes: Theorem 1.12	all
$\mathrm{SO}(3)$	Lecture 11	Lecture 11

Table 1.1: The uphill ledger. The group $(\mathbb{R}, +)$ sits in the framework as Lecture 8's unipotent 2×2 matrices; being an affine line, it is convex, and straight-line homotopies contract every loop.

Table 1.1 records the season's accounts. The $\mathrm{U}(1)$ condition unpacks pleasantly: a representation π of $\mathfrak{u}(1)$ is the single operator $A = \pi(\mathrm{i})$, the constraint $e^{2\pi A} = \mathbb{1}$ holds precisely when A is diagonalisable with every eigenvalue in $\mathrm{i}\mathbb{Z}$, and on each eigenline the integrated representation is a χ_n — Theorem 1.5, re-derived one eigenvalue at a time. The $\mathrm{SO}(3)$ row waits for Lecture 11, which will compute the loops of $\mathrm{SO}(3)$ by way of the adjoint map of Section 1.3. And the $\mathrm{SU}(2)$ row is a blank cheque — every representation of the algebra integrates to the group — which is the licence Lecture 10 spends lavishly.

1.5 Symmetries and conservation laws

Lecture 7 closed Part I with a promissory note: that the deepest consequence of continuous symmetry — conservation laws — would be the climax of Part II's opening movement. The note falls due. We pay it twice, once in each mechanics, and the two payments will turn out to be translations of a single sentence.

1.5.1 Classical mechanics: flows, brackets, Noether

Classical mechanics happens on phase space $\mathbb{R}^{2n}$, with coordinates $(q_1, \ldots, q_n, p_1, \ldots, p_n)$; observables are smooth functions, and the structure that organises them is the *Poisson* *bracket*

$$\{f, g\} = \sum_{a=1}^{n} \left(\frac{\partial f}{\partial q_a} \frac{\partial g}{\partial p_a} - \frac{\partial f}{\partial p_a} \frac{\partial g}{\partial q_a} \right). \tag{1.5}$$

It is bilinear and antisymmetric by inspection, and it satisfies the Jacobi identity by a direct, if tedious, computation: the smooth functions on phase space form a Lie algebra — the course's first infinite-dimensional one, met three sections after the definition stopped requiring matrices.

The bracket is not bookkeeping; it moves things. Every observable F generates a flow, the solution curves of $\dot{q}_a = \partial F/\partial p_a$, $\dot{p}_a = -\partial F/\partial q_a$, and along this flow every observable g changes at the rate

$$\frac{\mathrm{d}g}{\mathrm{d}s} = \{g, F\},$$

by the chain rule. Hamilton's equations are the special case $F = H$ with s the time: *dynamics is the flow of the energy*. Symmetry and dynamics are thus instances of one construction, and Noether's theorem is the observation that the construction is symmetric in its two arguments.

Theorem 1.18 (Noether's theorem, Hamiltonian form)**.** *For observables H and F on phase space, the following are equivalent:*

(i) H is constant along the flow of F (the flow of F is a symmetry of the dynamics);

(ii) $\{H, F\} = 0$;

(iii) F is constant along the flow of H (the quantity F is conserved in time).

Proof. Along the flow of F, $\mathrm{d}H/\mathrm{d}s = \{H, F\}$; along the flow of H, $\mathrm{d}F/\mathrm{d}t = \{F, H\} = -\{H, F\}$. Since a flow line passes through every point, either derivative vanishes identically precisely when the function $\{H, F\}$ does. ∎

Noether reciprocity — the symmetry preserves the energy if and only if the energy's own flow preserves the symmetry's generator — is nothing but the antisymmetry of the bracket. (Noether's own theorem is the Lagrangian, variational statement, of wider scope; the Hamiltonian form above is the version this course's structures see, and the one quantum mechanics will transcribe.)

The standard examples now compute themselves. The momentum p_x generates the flow $q_x \mapsto q_x + s$ with everything else fixed — spatial translation — so momentum is conserved exactly when H does not depend on q_x. The angular momentum $\ell_z = q_x p_y - q_y p_x$ generates the flow $\dot{q}_x = -q_y$, $\dot{q}_y = q_x$, $\dot{p}_x = -p_y$, $\dot{p}_y = p_x$, which rotates positions and momenta through the same angle about the third axis; rotation-invariant Hamiltonians conserve it. The three components close into a familiar shape,

$$\{\ell_a, \ell_b\} = \varepsilon_{abc}\, \ell_c\ : \tag{1.6}$$

the Poisson algebra of angular momenta is a copy of $\mathfrak{so}(3)$, realised on functions rather than matrices — Lecture 8's structure constants, found in the wild. And the ledger of Lecture 7 balances: a *finite* symmetry group contains no one-parameter flow to differentiate, which is why Part I, for all its selection rules, produced not a single conservation law. Each Hamiltonian one-parameter symmetry comes with a conserved generator. The generator is not unique: if F' generates the same flow, Hamilton's equations give $\partial(F - F')/\partial q_a = \partial(F - F')/\partial p_a = 0$, so $F - F'$ is constant on each connected component of phase space. Thus it is determined only up to an additive constant (one global constant on the phase space $\mathbb{R}^{2n}$ used here). That is the debt, repaid with interest.

1.5.2 Quantum mechanics: Stone's theorem and conservation

Quantum symmetries are unitary representations on the system's Hilbert space $\mathcal{H}$, and $\mathcal{H}$ is now allowed to be infinite dimensional — it must be, for the representations on wavefunctions that physics cares most about. Infinite dimensions force one decision that finite dimensions never posed: what should continuity mean?

Definition 1.19 (Strong continuity)**.** A *one-parameter unitary group* on a Hilbert space $\mathcal{H}$ is a map U from $\mathbb{R}$ to the unitary operators with $U(s+t) = U(s)U(t)$; it is *strongly continuous* if $t \mapsto U(t)\psi$ is continuous for every fixed $\psi \in \mathcal{H}$.

Demanding more — continuity in the operator norm — would exclude the most important examples. Take the translations $(T_a\psi)(x) = \psi(x-a)$ on $L^2(\mathbb{R})$ and any unit-norm bump ψ supported in an interval shorter than $|a|$: then $T_a\psi$ and ψ have disjoint supports, are orthogonal, and $\|T_a\psi - \psi\| = \sqrt{2}$. Hence $\|T_a - \mathbb{1}\| \geq \sqrt{2}$ for *every* $a \neq 0$: in the norm sense the translation group is nowhere continuous. Yet $\|T_a\psi - \psi\| \to 0$ as $a \to 0$ for each fixed ψ — continuity of translation in the mean, a standard fact of the students' analysis course, proved by approximating ψ by continuous functions of compact support. Strong continuity is the physically correct notion, and it is exactly what the fundamental theorem requires.

Theorem 1.20 (Stone's theorem; imported)**.** *Let U be a strongly continuous one-parameter unitary group on $\mathcal{H}$. Then there is a unique self-adjoint operator A — in general unbounded, defined on the dense domain of those ψ for which the limit exists — with*

$$A\psi = \mathrm{i} \lim_{t\to 0} \frac{U(t)\psi - \psi}{t} \qquad \text{and} \qquad U(t) = e^{-\mathrm{i}tA} \text{ for all } t \in \mathbb{R}.$$

Conversely, every self-adjoint A generates such a group. (For unbounded A the exponential is supplied by the spectral calculus of the students' functional analysis course.)

Reference for the omitted proof: Reed–Simon, *Methods of Modern Mathematical Physics* I, Theorem VIII.8; its spectral-calculus input is the spectral theorem in Theorem VIII.4. These are point-of-use imports, not declared prerequisites and not proved below.

In finite dimensions we have already proved this: strong and norm continuity coincide there, Lecture 8's one-parameter theorem produces a generator X, unitarity forces X anti-Hermitian by Theorem 1.6(v), and $A = \mathrm{i}X$ is the Hermitian generator of the physicists' convention from Lecture 8. Stone's theorem is the statement that this finite-dimensional fact survives, domains and all, in ∞ dimensions; we import it and use it at the level of care our functional analysis supports.

Its first consequence reframes dynamics itself. The Schrödinger equation $\mathrm{i}\dot{\psi} = H\psi$ integrates to $\psi(t) = U(t)\psi(0)$ with $U(t) = e^{-\mathrm{i}tH}$: a strongly continuous unitary representation of the group $(\mathbb{R}, +)$, whose Stone generator is the Hamiltonian. Conversely, by Stone, *any* continuous unitary dynamics has a Hamiltonian. Time evolution is not *like* a representation; it is one.

Example 1.21 (Translations and momentum)**.** The translations T_a above form a strongly continuous unitary representation of $(\mathbb{R}, +)$ on $L^2(\mathbb{R})$. On smooth compactly supported ψ the generator computes by the chain rule, $\mathrm{i}\,\frac{\mathrm{d}}{\mathrm{d}a}\psi(x-a)\big|_{a=0} = -\mathrm{i}\,\psi'(x)$, so

$$T_a = e^{-\mathrm{i}aP}, \qquad P = -\mathrm{i}\,\frac{\mathrm{d}}{\mathrm{d}x}. \tag{1.7}$$

The momentum operator of quantum mechanics is not a postulate: it *is* the generator of spatial translations. (That P is genuinely self-adjoint on its natural Sobolev domain, and is the closure of its restriction to standard smooth cores, is imported from the same spectral/ Stone theory.) Restoring units, $T_a = e^{-\mathrm{i}aP/\hbar}$ with $P = -\mathrm{i}\hbar\, \mathrm{d}/\mathrm{d}x$: Planck's constant is the exchange rate between the geometer's parameter, a length, and the physicist's observable, a momentum.

Example 1.22 (Rotations and angular momentum)**.** On $L^2(\mathbb{R}^3)$, Part I's function action $(\Pi(R)\psi)(x) = \psi(R^{-1}x)$ is a unitary representation of SO(3) — unitary because rotations preserve volume. For rotations about the third axis, $R = e^{\theta L_3}$, the chain rule on smooth ψ gives $\frac{\mathrm{d}}{\mathrm{d}\theta}\psi(e^{-\theta L_3}x)\big|_0 = (x_2\partial_1 - x_1\partial_2)\psi$, so with the physics dictionary $L_z = \mathrm{i}\,\pi(L_3)$ of Lecture 8,

$$\Pi\big(e^{\theta L_3}\big) = e^{-\mathrm{i}\theta L_z}, \qquad L_z = -\mathrm{i}\,(x_1\partial_2 - x_2\partial_1) = -\mathrm{i}\,\frac{\partial}{\partial\varphi}, \tag{1.8}$$

φ the azimuthal angle. Two reunions take place in this formula. First, the orbital angular momentum operator of every quantum mechanics course is here *derived*, not declared: it is what the geometry of rotations does to wavefunctions, differentiated. Second, its eigenfunctions $e^{\mathrm{i}m\varphi}$ must be single-valued on the circle $\varphi \equiv \varphi + 2\pi$, forcing $m \in \mathbb{Z}$ — and this is Theorem 1.5 again, because rotations about a fixed axis form a copy of U(1). Orbital angular momentum is integer-quantised *by topology*. Where, then, can half-integer angular momentum live? Not on functions on space: anything built from the function action represents U(1) along each axis, and the circle has spoken. Lecture 10 builds the half-integers' home; Lecture 11 explains the address.

The general theorem assembles these pieces. We state and prove it in finite dimensions. An infinite-dimensional version needs strong commutation of the relevant spectral measures and common invariant domains for products and commutators; that operator-theoretic result is imported only where later lectures invoke it, not proved by the matrix calculation below.

Theorem 1.23 (Noether's theorem, quantum form)**.** *Let G be a connected matrix Lie group, Π a unitary representation of G on a finite-dimensional $\mathcal{H}$, and $U(t) = e^{-\mathrm{i}tH}$ a quantum dynamics. The following are equivalent:*

(i) the symmetry commutes with the dynamics: $\Pi(g)\,U(t) = U(t)\,\Pi(g)$ for all $g \in G$, $t \in \mathbb{R}$;

(ii) $[H, \Pi(g)] = 0$ for all $g \in G$;

(iii) $[H, \pi(X)] = 0$ for all $X \in \mathfrak{g}$.

When they hold, every Hermitian generator $A = \mathrm{i}\,\pi(X)$ is conserved: $\langle\psi(t), A\,\psi(t)\rangle$ is constant in time for every solution $\psi(t) = U(t)\psi$ of the Schrödinger equation.

Proof. (i)$\Leftrightarrow$(ii): differentiating $\Pi(g)U(t)\Pi(g)^{-1} = U(t)$ at $t = 0$ gives $\Pi(g)H\Pi(g)^{-1} = H$; conversely, if $\Pi(g)$ commutes with H it commutes with every term of the exponential series for $U(t)$. (ii)$\Leftrightarrow$(iii): condition (ii) says exactly that H intertwines Π with itself, condition (iii) that H intertwines π with itself; over connected G these are equivalent by Theorem 1.6(iv). Conservation: with $A = \mathrm{i}\,\pi(X)$, condition (iii) gives $[H, A] = 0$, and

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\langle\psi(t), A\,\psi(t)\rangle = \langle -\mathrm{i}H\psi(t), A\,\psi(t)\rangle + \langle\psi(t), A\,(-\mathrm{i}H)\psi(t)\rangle = \mathrm{i}\,\langle\psi(t), [H, A]\,\psi(t)\rangle = 0. \qquad \blacksquare$$

The same computation, run for an arbitrary observable in the Heisenberg picture $A(t) = U(t)^\dagger A\,U(t)$, gives $\dot{A} = \mathrm{i}\,[H, A]$ — the quantum mirror of $\dot{f} = \{f, H\}$. Indeed the whole subsection has been a translation exercise, and the dictionary deserves a table.

classical mechanics	quantum mechanics
observable: smooth f	observable: self-adjoint A
bracket $\{f, g\}$	bracket $-\mathrm{i}\,[A, B]$
symmetry: the flow of F	symmetry: the group $e^{-\mathrm{i}sA}$
evolution $\dot{f} = \{f, H\}$	evolution $\dot{A} = \mathrm{i}\,[H, A]$
invariance $\{H, F\} = 0$	invariance $[H, A] = 0$
conserved: F along the motion	conserved: $\langle\psi, A\psi\rangle$ in time

Table 1.2: The Noether dictionary at $\hbar = 1$. The angular momenta obey it to the letter: compare (1.6) with $[L_a, L_b] = \mathrm{i}\,\varepsilon_{abc}L_c$. Lecture 13, Section *No finite-dimensional escape, and the Schrödinger representation*, returns to the canonical pair: its finite-dimensional trace and bounded-CCR obstructions force unbounded generators on a common core, whose bounded Weyl form is the rigorous replacement. This is not a promise of a general quantisation theorem for all classical observables.

Remark 1.24 (Degeneracy, continued)**.** Condition (ii) has a structural consequence that Part I trained us to expect. If $[H, \Pi(g)] = 0$ and $H\psi = E\psi$, then $H\,\Pi(g)\psi = \Pi(g)H\psi = E\,\Pi(g)\psi$: every energy eigenspace is invariant, so *energy levels carry representations of the symmetry group*. On a finite-dimensional complex isotypic eigenspace the same reduced-block rule applies: if an eigenvalue of the multiplicity-space matrix has multiplicity q, its total degeneracy is $q\dim V^{(i)}$. The irrep dimension is the symmetry-enforced multiplet size, not an assertion of exact total degeneracy. Equal eigenvalues from unrelated reduced blocks are accidental; conjugate blocks related by a real or antiunitary structure are forced to agree, as Lecture 6 explains. The $(2j{+}1)$-fold multiplets of atomic physics are already on their way: Lectures 10 and 11 deliver them.

The traffic between groups and algebras now runs in both directions, and the toll is computed: downhill is free over connected groups, uphill is free over simply connected ones, and the circle's winding is the model of what can go wrong. The physics has meanwhile assigned the algebra its profession — generators are observables, conservation is commutation with H, and quantisation can be the group's topology showing through. Everything is in place for the most consequential calculation of the course: the representations of $\mathfrak{su}(2)$.

Remark 1.25 (Computer algebra)**.** Notebook 9 verifies $\mathrm{Ad}(e^X) = e^{\mathrm{ad}_X}$ symbolically on $\mathfrak{su}(2)$; follows $e^{\mathrm{i}\lambda\theta}$ once around the circle for $\lambda = 1,\ \frac{1}{2},\ \sqrt{2}$ and plots the end-of-loop mismatch — the monodromy of Remark 1.16 made visible; compares partial sums of (1.4) against $\log(e^X e^Y)$ for random small $X, Y \in \mathfrak{sl}(2, \mathbb{R})$; and computes winding numbers by numerically lifting sampled loops.

Summary.

- *Differentiation.* Over a connected group, a representation is uniquely determined by its Lie-algebra representation under differentiation (Theorem 1.6); the irreducibles of $\mathrm{U}(1)$ are $\chi_n(e^{\mathrm{i}\theta}) = e^{\mathrm{i}n\theta}$, $n \in \mathbb{Z}$ (Theorem 1.5).
- *Adjoint.* $\mathrm{Ad}_g(X) = gXg^{-1}$ has derivative $\mathrm{ad}_X = [X, \cdot]$ (Definition 1.7, Proposition 1.8).
- *Integration and descent.* Simple connectivity guarantees that every Lie-algebra representation integrates uniquely (Theorem 1.15). For a connected group G, integrate on its universal cover $\widetilde{G}$ and descend exactly when the covering kernel acts trivially

(Remark 1.16); this recovers the integer weights of U(1).

- *Baker–Campbell–Hausdorff.* The product $e^X e^Y = e^{X+Y+\frac{1}{2}[X,Y]+\cdots}$ is determined by the bracket (Theorem 1.14).
- *Noether.* A Hamiltonian one-parameter symmetry has a conserved generator, defined up to an additive constant on each connected phase-space component (Theorem 1.18); quantum-mechanically the corresponding statement follows via Stone's theorem (Theorem 1.20, Theorem 1.23).

Tutorial. Exercise Sheet 9.

References

Hall §3.5, Ch. 4 (start), Ch. 5, especially §5.8 (BCH, lifting, and universal covers, as further reading); Woit Ch. 1–2 (quantum axioms and unitary representations) and the chapters on Fourier analysis and the free particle (Stone in action); Stillwell (simple connectivity); Osborn's notes (Noether).

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